Unknown-Word Handling in HMM POS Tagging

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Motivation

Currently, natural language processing is actively developing, and the task of partial speech tagging (POS tagging) remains one of the most fundamental and in-demand tasks. This topic is particularly relevant in the real world, where there is often a lack of labeled data and models face a large number of unknown words.
Therefore, studying effective strategies for handling unknown words in sequential-type models, such as Hidden Markov Models, is of great practical and scientific interest. This work aims to compare various methods for handling unknown words in order to improve the robustness and accuracy of POS tagging with limited training data.

Experimental Setup

In this work, experiments were conducted on the Brown Corpus, which contains 12 categories of parts of speech. The total volume of the corpus is 57,340 sentences. To simulate the conditions of limited labeled data, subsamples of 5%, 10%, 20%, 50%, and 100% were created from the training set.

A first-order hidden Markov model (HMM) was trained on each subsample. The Viterbi algorithm was used to find the most probable tag sequence.

Methodology

To handle unknown (out-of-vocabulary) words, five different strategies were implemented and compared:
Uniform: Assigns equal probability to all 12 tags for any unknown word (baseline approach).
Suffix-based: Uses suffixes of lengths 1 to 5 characters. Tag distributions are learned from the training data for each suffix, with longer suffixes having higher priority.
Pseudo-word Classes: Maps unknown words into linguistically motivated categories such as NUM (numbers), ALLCAPS (fully capitalized), CAPITALIZED (title case), HAS_HYPHEN (containing hyphen), and MIXED_DIGIT (containing digits and letters).
Character N-gram Backoff: Utilizes character prefixes and suffixes (n=1 to 3) as features to estimate the probability distribution over tags.
Hybrid: A combined strategy that applies the methods in the following order of priority: Pseudo-word Classes → Suffix-based → Character N-gram → Uniform (as a fallback).
Add-k smoothing options: Returns the global tag distribution estimated from the training data, smoothed additively using the k parameter

Pipeline

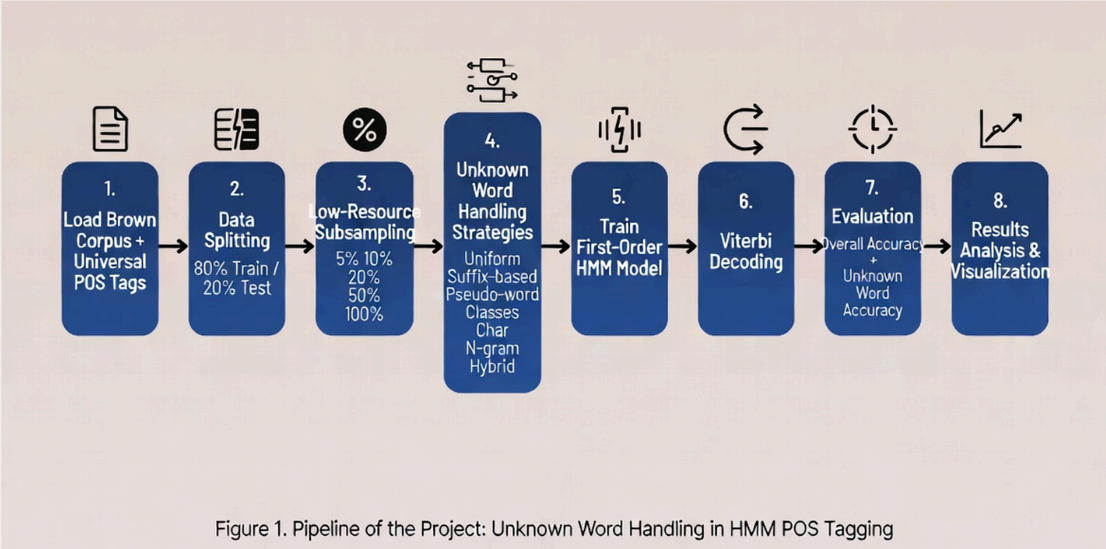
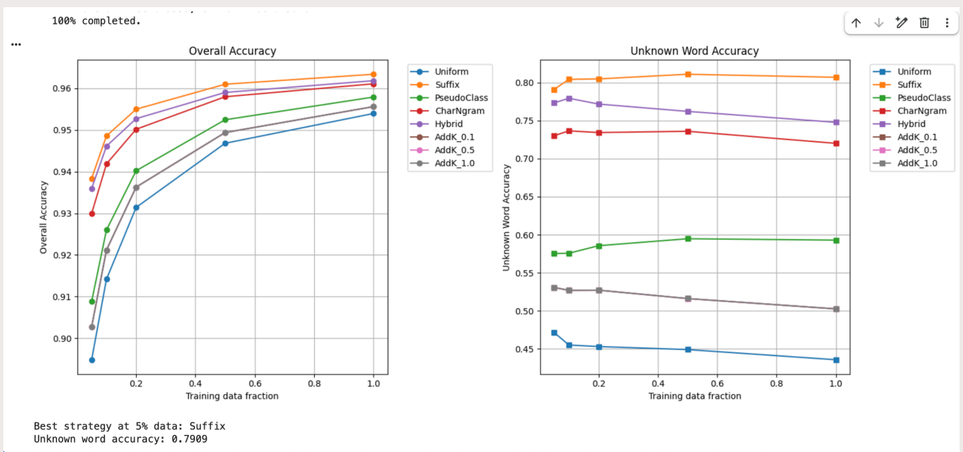


Figure 1. Pipeline of the Project: Unknown Word Handling in HMM POS Tagging

Results

The project successfully evaluated different unknown-word handling strategies in HMM-based POS tagging under limited annotated data conditions. Among all tested approaches, the Suffix-based strategy demonstrated the most significant improvement in tagging robustness, especially when only 5–20% of the training data was available. The Hybrid strategy also showed strong performance by combining multiple methods and achieved the best balance between overall accuracy and unknown word accuracy in extremely low-resource settings. Overall, when annotated data is limited, the Suffix-based heuristic is recommended as the most effective and stable unknown-word handling strategy for HMM POS tagging.



Conclusion

In this work, a hidden Markov model (HMM) was implemented for part-of-speech (POS) tagging, and various strategies for handling unknown words were compared. Experiments were conducted with limited labeled data, using subsamples from 5% to 100% of the Brown corpus training set.
The results showed that strategies that take into account the morphological structure of words significantly outperform the basic approach. The suffix heuristic and the hybrid strategy demonstrated the best results in terms of handling unknown words.